

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

26th November 2019

Subject SA1 – Health and Care

Time allowed: 3 Hours 15 Minutes (14.45 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** You work in a big multinational Health & Care Insurance company in a senior position. The Government of a developing country in Asia is concerned about the problems due to changes in Socio-Economic factors and particularly the problems of increasing need of care at old ages post retirement. The Government Health & family Welfare Department officials have primarily agreed to have a Long Term Care Insurance product for the citizens from a reputed Insurance Company. Some form of budgetary support will be provided by the Government in the form of cost sharing. Your Company is selected by the Government as one such provider. The Scheme is at the discussion stage and not finalized yet. The Government has called you as the representative of your employer to discuss it in further details. It came in discussion that the benefits may be provided in the form of care required or in cash which is either fixed or increases at 3% p.a. (3% p.a. is the long term inflation forecast for the Country). Also, there will be pre-funded and Immediate needs versions of the Long Term Care product. The concerned minister is of the opinion that fixed benefits (increasing at 3% p.a. post retirement) will be the appropriate form.
- i)** Explain the advantages and disadvantages of the fixed benefit (increasing at 3% p.a. post retirement) product both to the insured and the insurer. (8)
 - ii)** Explain the differences between a pre-funded and an immediate needs long-term care insurance policy. Also, describe the advantages and disadvantages for the insured of each of these designs. (16)
- Finally, the scheme is in operation and your company is operating in a large portion of the country with close interactions with the Government. You are Chief Actuary of the Scheme & you have entered in to a reinsurance arrangement with a large reinsurer. About 20% of your portfolio is reinsured. The premium rates may be changed only with the approval of the Government. The Government provides support in distribution of the product. It has achieved a healthy volume of business.
- iii)** Explain why you would want to monitor the experience of your company & your Reinsurer and profitability of this business on an ongoing basis. (10)
 - iv)** Describe suitable asset classes for the company's investments to support the liabilities arising from the block of business written. (14)
- After 2 years, your business has grown substantially with pre-funded and immediate need long term care products in the country. You have performed the annual analysis of the long term business. It is found that due to negative operating variances with respect to expenses and morbidity, their contributions towards change in Embedded Value are negative. The contribution of morbidity was positive last year but was negative for expenses. The CFO of the company has asked your opinion about initiating premium rate change discussion with Government and about revising the assumptions used in EV calculation.
- v)** Draft a reply explaining the negative operating variances & various assumptions used in EV calculations. Your reply should cover how changes in assumptions may lead to the premium change discussion with the Government. (12)
- [60]**

- Q.2)** The shareholders of a large general insurance company and a stand-alone health insurance (SAHI) company intend to merge together. Post the merger, the SAHI will cease to exist and the merged entity will have the same brand name as the GI company.

The General Insurance Company currently doesn't underwrite significant amount of health business while the stand alone health insurance company is one of the leading health insurers of the country.

- i)** Discuss the possible motives for mergers and acquisition. (12)
- ii)** The deal will need to be approved by the regulatory authority. List the information that may form part of the proposal. (6)
- iii)** Discuss the issues to be considered by the merged entity regarding the following:
 - a)** Products (3)
 - b)** Employees (3)
 - c)** Systems / IT infrastructure (3)
 - d)** Distribution channels (3)
- iv)** You are a consulting actuary and have been hired by the shareholders of the general insurance company to advise them on this deal. Explain the methodology to compute the appraisal value of the Company. (10)

[40]
